Image and Vision Computing 114 (2021) 104266

Contents lists available at ScienceDirect

Image and Vision Computing

journal homepage: www.elsevier.com/locate/imavis

A bioinspired retinal neural network for accurately extracting small-target motion information in cluttered backgrounds

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ARTICLE INFO

Article history:
Received 12 July 2021
Accepted 30 July 2021
Available online 6 August 2021

Keywords:
Bioinspiration
Small-target motion detector
Robotic visual perception
Spatiotemporal energy model

ABSTRACT

Robust and accurate detection of small moving targets in cluttered moving backgrounds is a significant and challenging problem for robotic visual systems to perform search and tracking tasks. Inspired by the neural circuitry of elementary motion vision in the mammalian retina, this paper proposes a bioinspired retinal neural network based on a new neurodynamics-based temporal filtering and multiform 2-D spatial Gabor filtering. This model can estimate motion direction accurately via only two perpendicular spatiotemporal filtering signals, and respond to small targets of different sizes and velocities through adjusting the dendrite field size of spatial filter. Meanwhile, an algorithm of directionally selective inhibition is proposed to suppress the target-like features in the moving background, which can reduce the influence of background motion effectively. Extensive synthetic and real-data experiments show that the proposed model works stably for small targets of a wider size and velocity range, and has better detection performance than other bioinspired models. Additionally, it can also extract the information of motion direction and motion energy accurately and rapidly.

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1. Introduction

Rapidly detecting small moving targets in the cluttered background and judging their motion directions, is very important for most animals to track prey and escape predators. It is also a significant and challenging problem for robotic visual systems to perform search and rescue, traffic monitoring, and any task that requires detection of small moving targets. However, it is difficult to detect these kind of targets since most shape, color, texture and structure information is lost in few pixels. Meanwhile, these targets are easily buried in the background and are indistinguishable from noise such that it becomes difficult to detect small targets under such low signal-to-clutter ratio (SCR) backgrounds. Thus, the motivation of this paper is to detect small moving targets quickly and accurately in the natural cluttered environment, and extract more motion information like motion direction and energy at the same time.

For motion detection, most of conventional computer-vision solutions are for normal-sized objects like pedestrians and vehicles, including background subtraction [1,2], temporal differencing [3,4], statistical model [5,6] and optical flow [7,8]. Recent years have also seen many deep learning-based target detection methods, such as CNN framework [9–11] and YOLO framework [12–14]. While most of conventional solutions to small-target detection focus on infrared images [15–17]. For instance, Wang et al. [18] built an infrared dim target detector using total variation regulation and principal component pursuit to supress background noise and improve detection performance. However, these infrared-based approaches rely on the strong temperature difference between the background and the targets. For natural images, Zhu et al. [19] recently proposed a CNN-based method called YOLOv3-SOD for detecting small targets. But most of deep learning-based methods usually need a lot of training data and manual annotation for some specific targets instead of any small moving objects.

Recently, many bioinspired models have emerged to perform this kind of motion detection. In nature, insects have excellent ability to search for moving features in the distance, which derives from a special class of small target motion-detecting neurons (STMDs) [20,21]. Inspired by visual system in insects, some studies recently have developed several efficient models for elementary visual perception in cluttered environments. For instance, the study in [22] firstly proposed a computational model for target discrimination, which detected most targets from the background successfully. After that, [23–25] improved this model further by using a facilitation mechanism and motion direction detection system, but motion direction is only divided into right/up and left/down. Similarly, [26] built a computational model of directionally selective small target motion detection (DSTMD) to not only perform target discrimination, but also perceive the eight directions of movement roughly. However, these models generally suffer from some performance loss of detection since they were just sensitive

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https://doi.org/10.1016/j.imavis.2021.104266
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to a small range of target sizes and velocities and could not suppress the movement of background effectively. The study [27] improved the performance of DSTMD by using directional contrast information to discriminate small targets from fake features of background. This approach actually requires a prior determination of the motion direction of background instead of automatic detection.

The visual acuity of mammalian vision is generally higher [28], but the temporal resolution is relatively low [29] due to more complicated top-down interaction. But for low-level visual processing, both retinas actually have common neural circuit design to achieve primary motion perception [30,31]. Thus, it is attractive and significant to model mammalian low-level vision, and further develop higher-level visual cognition system for general machine intellegence. Currently, bioinspired studies for modeling mammalian motion vision are relatively few. A feasible solution is to use spatiotemporal energy models. For instance, Adelson and Bergen [32] used Gabor filters and a temporal band-pass structure to achieve elementary perception of motion, and some studies used this model to calculate the optical flow [33]. However, such approaches generally use infinite impulse response (IIR) filters to computing the output signals, where a phase delay is usually encounted. Another bioinspired model is the ViSTARS based on neural dynamics [34,35], which demonstrates how primates use motion information to segment objects and compute heading from optic flow and obstacle avoidance. This model is biologically plausible but with large number of dynamic parameters, which is difficult to find suitable parameters for adapting to the cluttered and dynamic environments. Meanwhile, all aforementioned approaches are mainly used for wide-field motion perception rather than small-field motion detection.

Inspired by the neural circuitry of elementary motion vision in the mammalian retina, this paper proposes a bioinspired retinal neural network (BRNN) that can not only detect small moving targets in cluttered backgrounds, but also estimate motion direction and motion energy of targets accurately. These advantages mainly stem from the following contributions.

1. a novel spatiotemporal energy model is proposed based on a neurodynamics-based temporal filtering and multiform 2-D spatial Gabor filtering. The motion direction can be estimated accurately by only two perpendicular direction-selective spatiotemporal filtering channels. The optimal sensing size and velocity can be mediated by adjusting the dendrite field size of antagonistic center-surround spatial filtering.
2. An algorithm of directionally selective inhibition is proposed to suppress the target-like features in the background. If a large number of targets appear suddenly in similar directions, these target-like features will be regarded as interference targets, then the spatial filtering in these directions will be inhibited. The method can reduce the influence of background motion effectively.

The rest of this paper is organized as follows. Section 2 introduces the neural mechanisms of visual motion processing in the mammalian retina. Section 3 provides a detailed description of BRNN. Section 4 describes the experimental setup and discusses the results of motion perception in synthetic and real scenarios respectively. Section 5 concludes this paper.

2. Visual motion processing in the mammalian retina

The mammalian retinas generally have a layered structure that contains five principle cell types: photoreceptors, horizontal cells, bipolar cells, amacrine cells and ganglion cells [36], as shown in Fig. 1(a). Photoreceptor cells, including rods and cones, are mainly in charge of capturing the change of luminance. Next, the visual signal divides into two parallel channels. The ON channel responds to contrast increments and the OFF channel responds to contrast decrements [37]. Then fast and slow responses are created based on different temporal dynamics in bipolar cells [38,39], which are mainly responsible for temporal filtering. Meanwhile the bipolar cells are also affected by the inhibitory connections from the horizontal cells.

ON and OFF bipolar cells are further connected to the amacrine cells that perform spatial filtering. One type of them, called starburst amacrine cells (SACs), has been discovered to be sensitive to its preferred direction of motion [30]. Another class of them, calld wide-field amacrine cells (WACs) could endow the direction-selective circuit with an additional feature: size selectivity [41]. WACs contact bipolar cells and drive direct excitation and feedforward inhibition (through SACs) to ganglion cells, which allows the size selectivity to be adjusted. Finally, retinal ganglion cells (RGCs) process the visual information from the amacrine cells and implement elementary motion perception further. Among them, three groups of RGCs have been found to be sensitive to the motion direction, ON-OFF RGCs, ON RGCs and OFF JAM-B cells [30], where ON-OFF RGCs are able to respond to both ON and OFF edges moving in different directions with a broad range of speeds [42]. Meanwhile, these direction-selective ganglion cells (DSGCs) are able to encode the size information of objects under the modulation of WACs.

3. Bioinspired retinal neural network for motion detection

Based on above neural mechanisms, a computational neural network model is proposed in this section. This model mainly contains four layers (Fig. 1(b)): (1) photoreceptor cells capture the change of luminance. (2) Bipolar cells encode the visual information into ON and OFF pathways, and perform fast and slow temporal filtering. (3) SACs perform direction-selective spatial filtering and WACs perform antagonistic center-surround spatial filtering. (4) RGCs discriminate motion patterns through computing the motion energy based on spatiotemporal filtering information.

3.1. Photoreceptor cells

The first layer consists of photoreceptor cells that receive gray-scale visual stimuli and capture the change of luminance. These cells are arranged as a 2D matrix that is the same size as the input image.

$$P(x,y,t) = I(x,y,t) - I(x,y,t-1) + \sum_{i}^{N_p} p_i P(x,y,t-i) \tag{1}$$

where $P(x,y,t)$ represents the change of brightness corresponding to each pixel at time t, x and y are the abscissa and ordinate of the pixel respectively. $I(x,y,t)$ and $I(x,y,t-1)$ are the gray-scale value of two successive frames. Assume the change of luminance lasts for N_p number of frames, but fast decays with coefficient p_i. Herein, we define $p_i = (1+e^{ui})^{-1}$, and choose $u = 1$.

3.2. Bipolar cells

In the bipolar cells, the visual signal is split and processed separately in ON and OFF channels. The ON channel responds to contrast increments and OFF channel responds to contrast decrements. Herein, a 'half-wave' rectifier can be used to simulate the ON–OFF split as follows.

$$B_+(x,y,t) = \frac{1}{2}[|P(x,y,t)| + P(\mathrm{x},\mathrm{y},\mathrm{t})] \tag{2}$$

$$B_-(x,y,t) = \frac{1}{2}[|P(x,y,t)| - P(\mathrm{x},\mathrm{y},\mathrm{t})] \tag{3}$$

Meanwhile, the bipolar cells receive some inhibitory signals from the horizontal cells. Herein, difference of gaussians is used to model this process as a bandpass filter. The kernel function is the difference of two gaussians with different standard deviation

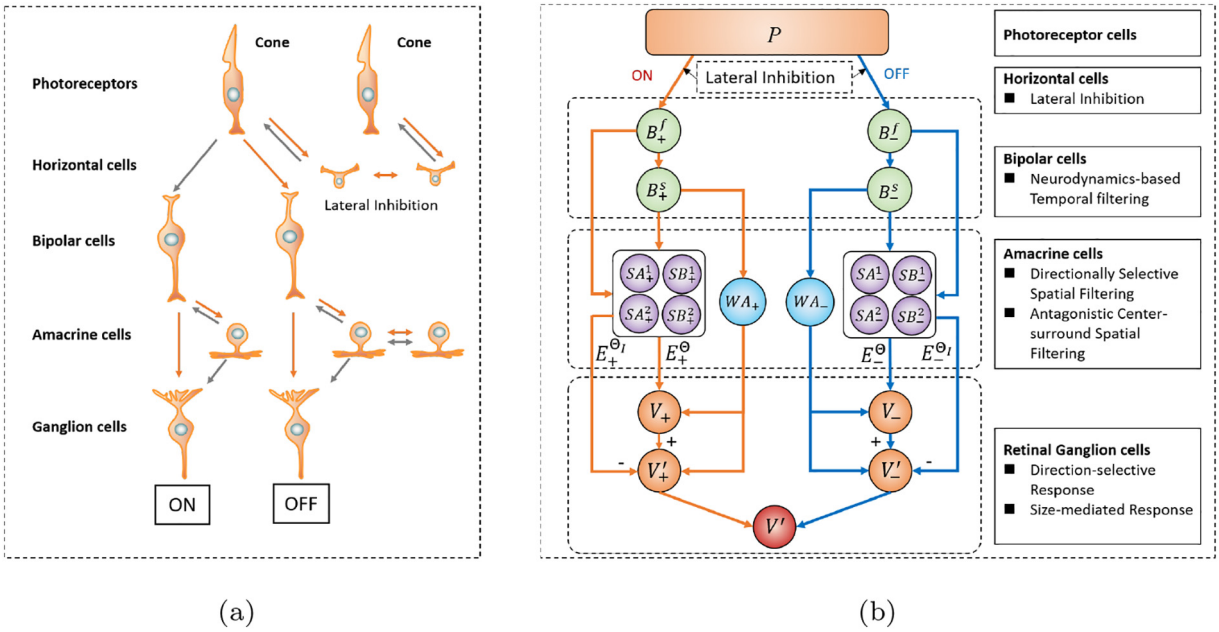

Fig. 1. The neural mechanism and computational model of visual motion processing in the mammalian retina. (a) The retinal circuitry for cone signals [40]. (b) The computational model of primary visual motion perception.

$$g_I(x,y;\sigma_1,\sigma_2) = \left(\frac{F}{\sqrt{2\pi}\sigma_1}e^{\frac{-(x^2+y^2)}{2\sigma_1^2}} - \frac{F}{\sqrt{2\pi}\sigma_2}e^{\frac{-(x^2+y^2)}{2\sigma_2^2}}\right) \tag{4}$$

where F is the gain coefficient, σ_1 and σ_2 are the standard deviation of two gaussians. The activations of the bipolar cells can be computed through convolving with this kernel function.

$$\begin{aligned} B_+^0(x,y,t) &= B_+(x,y,t)\otimes g_I \\ B_-^0(x,y,t) &= B_-(x,y,t)\otimes g_I \end{aligned} \tag{5}$$

where $B_+^0(x,y,t)$ and $B_-^0(x,y,t)$ are the neural activity of ON and OFF bipolar cell in the first layer respectively. $\otimes$ represents the convolution operation.

In order to compute the neural activities with different response speed, a new temporal filtering method is proposed based on cascades of leaky integrators. The structure of this filter consists of n-level cascade of leaky integrators, denoted as $\mathcal{L}(K,n,z_0)$. K is the gain of filter, n is the number of layers, and z_0 is the input signal. Specifically, nth section of leaky integrators obeys the following dynamic equation.

$$\tau\frac{dz_n}{dt} = -Az_n + Cz_{n-1} \tag{6}$$

where τ is a time constant of integration, A is a decay coefficient, and C is a coefficient of transmission. Finally, the output of temporal filtering is defined as the difference between two layers of neurons.

$$\mathcal{L}(K,n,z_0) = K(z_n - z_{n+m}) \tag{7}$$

where $m > 0$. In this paper, we choose $m = 1$.

Then, the fast and slow responses of ON and OFF bipolar cells are computed respectively as follows

$$\begin{aligned} B_+^f(x,y,t) &= \mathcal{L}\left(K,n_f,B_+^0\right) \\ B_+^s(x,y,t) &= \mathcal{L}\left(K,n_s,B_+^0\right) \\ B_-^f(x,y,t) &= \mathcal{L}\left(K,n_f,B_-^0\right) \\ B_-^s(x,y,t) &= \mathcal{L}\left(K,n_s,B_-^0\right) \end{aligned} \tag{8}$$

where $n_f < n_s$.

3.2.1. Analysis of the temporal filtering

The response of this model to an instantaneous unit impulse can be calculated simply. For the dynamic Eq. (6), if p is written for d/dt, the equation relating output z_n to input z_0 is

$$\left(p + \frac{A}{\tau}\right)^n z_n = \left(\frac{C}{\tau}\right) z_0 \tag{9}$$

When z_0 is unit impulse signal, the solution of above equation is

$$z_n = \left(\frac{C}{\tau}\right)^n \frac{t^{n-1}e^{\frac{-At}{\tau}}}{(n-1)!} \tag{10}$$

Since output of temporal filtering $\mathcal{L}$ is the difference between nth and $(n+m)$th layers of neurons, the unit impulse response of Eq. (7) is

$$\begin{aligned} \mathcal{L} &= K\left(\frac{C}{\tau}\right)^n \frac{t^{n-1}e^{\frac{-At}{\tau}}}{(n-1)!} - K\left(\frac{C}{\tau}\right)^{n+m} \frac{t^{n+m-1}e^{\frac{-At}{\tau}}}{(n+m-1)!} \\ &= K\left(\frac{C}{\tau}\right)^n t^{n-1}e^{\frac{-At}{\tau}}\left[\frac{1}{(n-1)!} - \frac{\left(\frac{Ct}{\tau}\right)^m}{(n+m-1)!}\right] \end{aligned} \tag{11}$$

Let $a = \frac{A}{\tau}$, $b = \frac{C}{\tau}$, then we have

$$\mathcal{L} = Ke^{-at}\left[\frac{b^n t^{n-1}}{(n-1)!} - \frac{b^{n+m}t^{n+m-1}}{(n+m-1)!}\right] \tag{12}$$

The derivative with respect to t is

$$\frac{d\mathcal{L}}{dt} = -Kae^{-at}\left[\frac{b^n t^{n-1}}{(n-1)!} - \frac{b^{n+m} t^{n+m-1}}{(n+m-1)!}\right] + Ke^{-at}\left[\frac{b^n t^{n-2}}{(n-2)!} - \frac{b^{n+m} t^{n+m-2}}{(n+m-2)!}\right] \tag{13}$$

Let $\frac{d\mathcal{L}}{dt} = 0$, we have

$$\left[\frac{b^n t^{n-2}}{(n-2)!} - \frac{b^{n+m} t^{n+m-2}}{(n+m-2)!}\right] - a\left[\frac{b^n t^{n-1}}{(n-1)!} - \frac{b^{n+m} t^{n+m-1}}{(n+m-1)!}\right] = 0 \tag{14}$$

Simplify this equation, there is

$$\frac{n-1-at}{(n-1)!} - \frac{b^m t^m (n+m-1-at)}{(n+m-1)!} = 0 \tag{15}$$

Then we have

$$ab^m (n-1)! t^{m+1} - b^m (n+m-1)(n-1)! t^m - a(n+m-1)! t + (n-1)(n+m-1)! = 0 \tag{16}$$

Especially when $m = 1$, we have

$$ab(n-1)! t^2 - (a+b)n! t + (n-1)n! = 0 \tag{17}$$

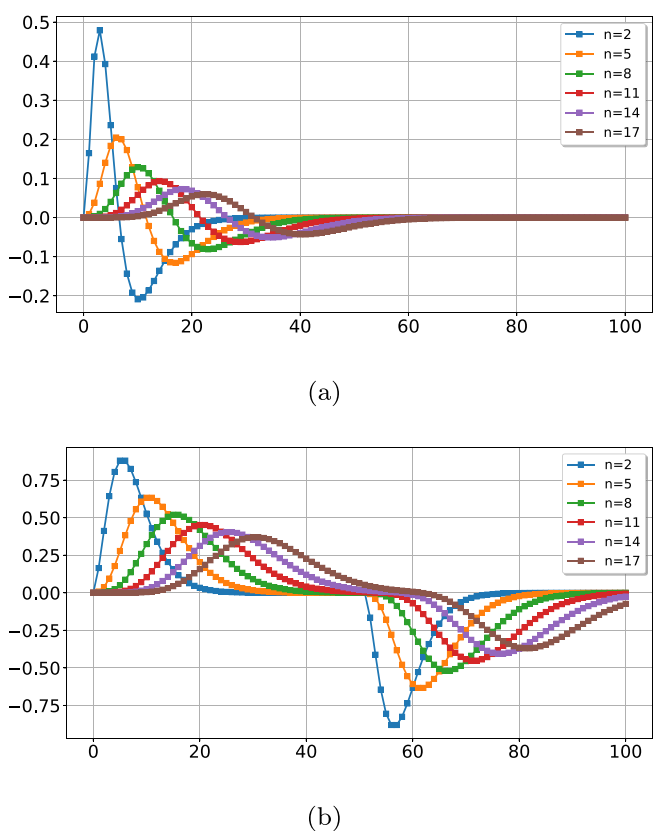

Fig. 2. The response characteristics of the neurodynamics-based temporal filter. (a) The unit impulse response of temporal filter at different n. (b) The step response of temporal filter at different n.

Since

$$\begin{aligned}\Delta &= [(a+b)n!]^2 - 4ab(n-1)! \cdot (n-1)n! \\ &= (an!)^2 + (bn!)^2 + 2ab(n!)^2 - 4ab(n-1)! \cdot (n-1)n! \\ &> (an!)^2 + (bn!)^2 + 2ab(n!)^2 - 4ab(n!)^2 \\ &= [(a-b)n!]^2 \geq 0\end{aligned} \tag{18}$$

there are two real roots.

$$t_1 = \frac{(a+b)n! - \sqrt{\Delta}}{2ab(n-1)!} \tag{19}$$

$$t_2 = \frac{(a+b)n! + \sqrt{\Delta}}{2ab(n-1)!} \tag{20}$$

The activity reaches a maximum at time t_1, and reaches a minimum at time t_2. When $A > C$ $(a > b)$, then $t_1 < \frac{n}{a}$ and $t_2 > \frac{n}{b}$. Otherwise when $A < C$ $(a < b)$, then $t_1 < \frac{n}{b}$ and $t_2 > \frac{n}{a}$.

We conduct a simple simulation to observe the characteristics of this neurodynamics-based temporal filter. The total simulation time is 100 steps, and each step is set as $\Delta t = 0.05s$. The gain of filter $K = 5$, the decay coefficient $A = 60$, the coefficient of transmission $C = 60$, and $\tau = 8$. The unit impulse response and step response of temporal filter at different n are shown in Fig. 2(a) and (b) respectively. Obversely, the response speed decreases with the increase of n. Meanwhile, the response amplitude is also related to the number of cascades n. As n increases, the response gradually weakens.

Actually, the above response is similar to the temporal impulse responses described in [32]. The filtering result is computed offline through convolving the signal history with the following kernel function.

$$f(t) = \frac{(kt)^n}{n!} e^{-kt} - \frac{(kt)^{n+2}}{(n+2)!} e^{-kt} \tag{21}$$

Although it can serve as plausible approximations to psychophysical data, it is not biologically plausible for the neural computation process. In addition, this method needs convolution along the time dimension, which usually suffers from typical phase delay or phase shift problem such that real-time performance is affected.

3.3. Amacrine cells

3.3.1. Direction-selective spatial filtering

SACs perform direction-selective spatial filtering. Frequency and orientation representations of Gabor filters are demonstrated to be similar to those of the human visual system. Thus, Gabor filters are used to perform spatial filtering of fast and slow bipolar neurons. The kernel function is

$$g_S(x, y; \lambda_S, \theta, \sigma_S) = \exp\left(-\frac{x'^2 + y'^2}{2\sigma_S^2}\right) \cos\left(\frac{2\pi x'}{\lambda_S} + \psi\right) \tag{22}$$

where $x' = x\cos(\theta) + y\sin(\theta)$, $y' = -x\sin(\theta) + y\cos(\theta)$. λ_S is wavelength of the sinusoidal factor, which is usually measured in pixels. θ represents the orientation of the Gabor function. ψ is the phase offset. σ_S is the standard deviation of the Gaussian envelope. The shape of Gabor kernel function at different orientations is shown in Fig. 3(a), where $\lambda_S = 2$, $\sigma_S = 3$, $\psi = 0$.

Herein, two gabor filters are used to process the visual information from the bipolar cells. The phase offset of the first filter g_1 is $\psi_1 = 0$, and the one of the second filter g_2 is $\psi_2 = \frac{\pi}{2}$. Then, for each orientation θ, the neural activities of the starburst amacrine cells are calculated in the following way.

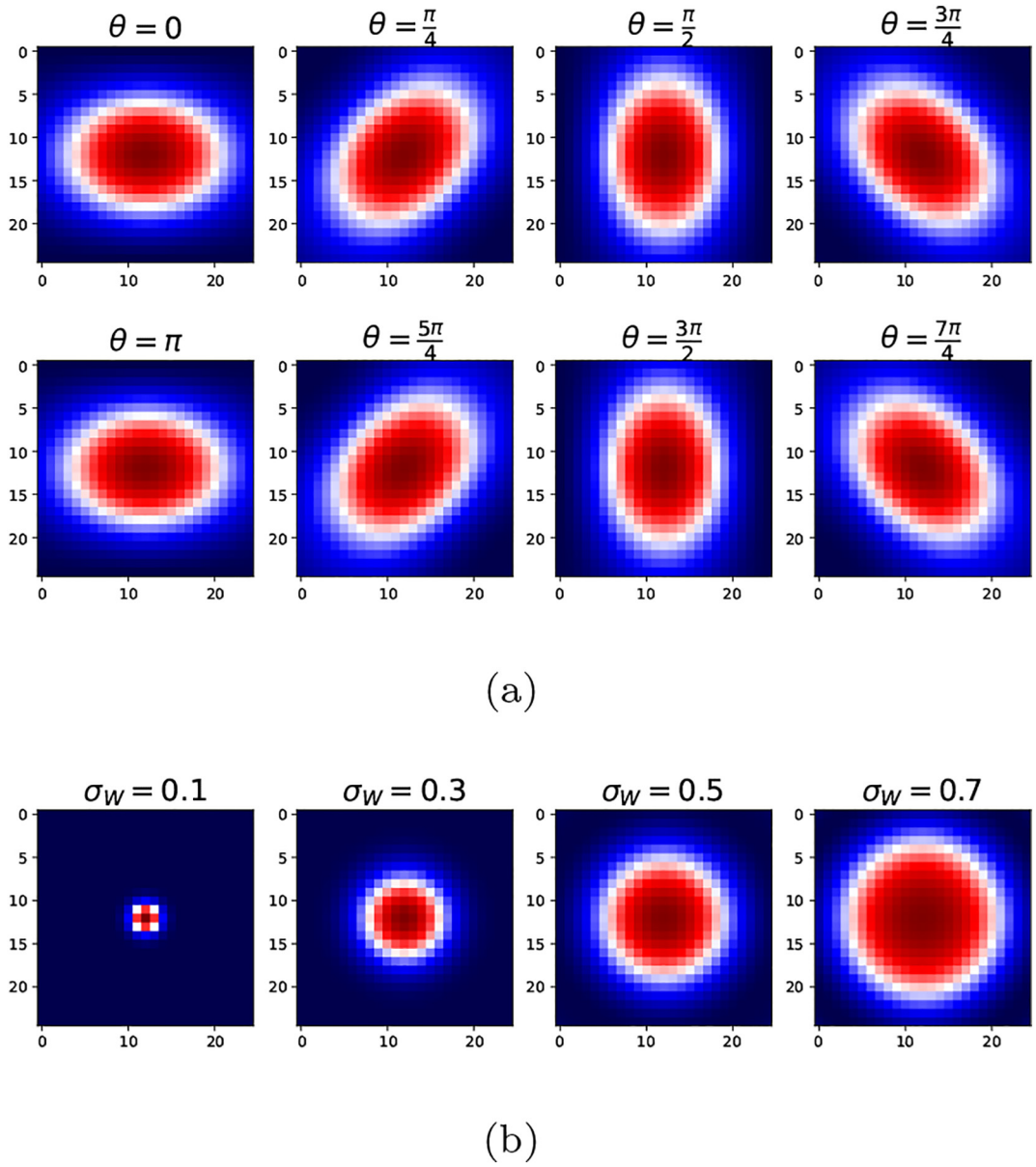

Fig. 3. The shape of spatial filters with different parameters. (a) The shape of direction-selective Gabor filters at different orientations. (b) The shape of antagonistic center-surround Gabor filters at different standard deviations.

$$\begin{aligned}
SA_+^1(x,y,t) &= B_+^s(x,y,t)\otimes g_1\\
SB_+^1(x,y,t) &= B_+^s(x,y,t)\otimes g_2\\
SA_+^2(x,y,t) &= B_+^f(x,y,t)\otimes g_1\\
SB_+^2(x,y,t) &= B_+^f(x,y,t)\otimes g_2\\
SA_-^1(x,y,t) &= B_-^s(x,y,t)\otimes g_1\\
SB_-^1(x,y,t) &= B_-^s(x,y,t)\otimes g_2\\
SA_-^2(x,y,t) &= B_-^f(x,y,t)\otimes g_1\\
SB_-^2(x,y,t) &= B_-^f(x,y,t)\otimes g_2
\end{aligned} \tag{23}$$

where $\otimes$ represents the convolution operation. Eight kinds of SACs are defined here for computing the motion energy in the ganglion cells.

3.3.2. Antagonistic center-surround spatial filtering

WACs mainly perform antagonistic center-surround spatial filtering to control size selectivity. Herein, a new kernel function is defined following the formation of Gabor filters. It can excite the center and inhibit the surrounding without direction selectivity.

$$g_W(x,y;\lambda_W,\sigma_W) = 2\exp\left(-\frac{x^2+y^2}{2\sigma_W^2}\right)\cos\left(\frac{2\pi x}{\lambda_W}\right)-1 \tag{24}$$

where λ_W is wavelength of the sinusoidal factor, σ_W is the standard deviation of the Gaussian envelope. The kernel size is denoted as $M_W \times M_W$. The balance of excitation/inhibition is mainly regulated by the kernel size M_W and the standard deviation σ_W, which plays a dominant role in controlling size selectivity. The shape of this kernel function at different σ_W is displayed in Fig. 3(b), where $M_W = 25$.

WACs contact bipolar cell terminals and generate mediated signals to control ganglion cell responses. Specifically, the ON and OFF mediated signals are obtained as follows.

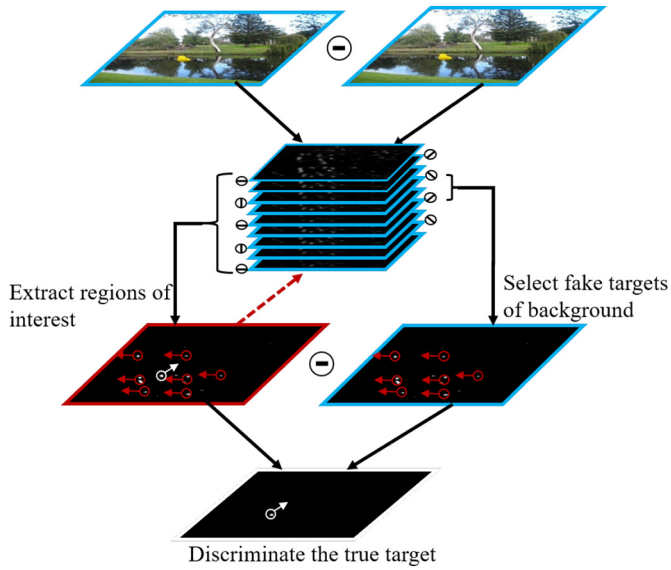

Fig. 4. The diagram of directionally selective inhibition.

$$\begin{aligned} WA_{+}(x,y,t) &= \left[\left[B^{s}_{+}(x,y,t)\right]^{+} \otimes g_{W}\right]^{+} \\ WA_{-}(x,y,t) &= \left[\left[B^{s}_{-}(x,y,t)\right]^{+} \otimes g_{W}\right]^{+} \end{aligned} \tag{25}$$

where $[\cdot]^{+} = \max(0,\cdot)$ represents a rectified linear unit. This process can boost center and inhibit surrounding response based on activations of bipolar cells.

3.4. Retinal ganglion cells

3.4.1. Motion direction estimation

In the motion direction sensing task, RGCs fuse the spatiotemporal filtering results of different phases and different speeds to calculate the motion energy of each pixel. According to the motion energy in each direction, then, the motion direction can be discriminated. Following the Hassenstein-Reichardt model [43] and Adelson-Bergen model [32], the motion energy at each filtering orientation θ can be computed as follows.

Table 1
The predefined parameters of BRNN model.

Name	Value	Name	Value	Name	Value
F	5.0	σ_1	1.0	σ_2	3.0
A	60.0	τ	5.0	Δt	0.05
K	5.0	n_f	2	λ_S	4
σ_S	0.3	–	–	–	–

$$\begin{aligned} E^{\theta}_{+}(x,y,t) &= SA^{1}_{+}(x,y,t)\cdot SB^{2}_{+}(x,y,t) \\ &\quad - SA^{2}_{+}(x,y,t)\cdot SB^{1}_{+}(x,y,t) \\ E^{\theta}_{-}(x,y,t) &= SA^{1}_{-}(x,y,t)\cdot SB^{2}_{-}(x,y,t) \\ &\quad - SA^{2}_{-}(x,y,t)\cdot SB^{1}_{-}(x,y,t) \end{aligned} \tag{26}$$

Additionally, there are several types of RGCs that recombine ON and OFF signals from SACs, such as ON–OFF direction-selective retinal ganglion cells. This type of cells responds to both contrast increments and decrements along a specific direction. Herein, ON–OFF motion energy is defined in a weighted summation of ON and OFF pathways.

$$E^{\theta}(x,y,t) = w_{+}E^{\theta}_{+}(x,y,t) + w_{-}E^{\theta}_{-}(x,y,t) \tag{27}$$

where w_{+} and w_{-} are weighting coefficients of ON and OFF signals. The motion direction can be estimated with only two mutually perpendicular spatial filtering processes, which are the motion energy at 0 and $\frac{\pi}{2}$ respectively.

$$\widehat{\varphi} = \arctan 2\left(E^{\frac{\pi}{2}}(x,y,t), E^{0}(x,y,t)\right) \tag{28}$$

3.4.2. Size-mediated motion detection with moving background Suppresion

WAC-mediated responses of RGCs are computed by combining the maximal motion energy and the mediated signals. Assume $E^{\Theta}(x,y,t)$ is the set of neural outputs at all filtering orientations Θ, $\Theta = \{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}\}$. Then activities of size-selective ganglion cells are determined in the following way.

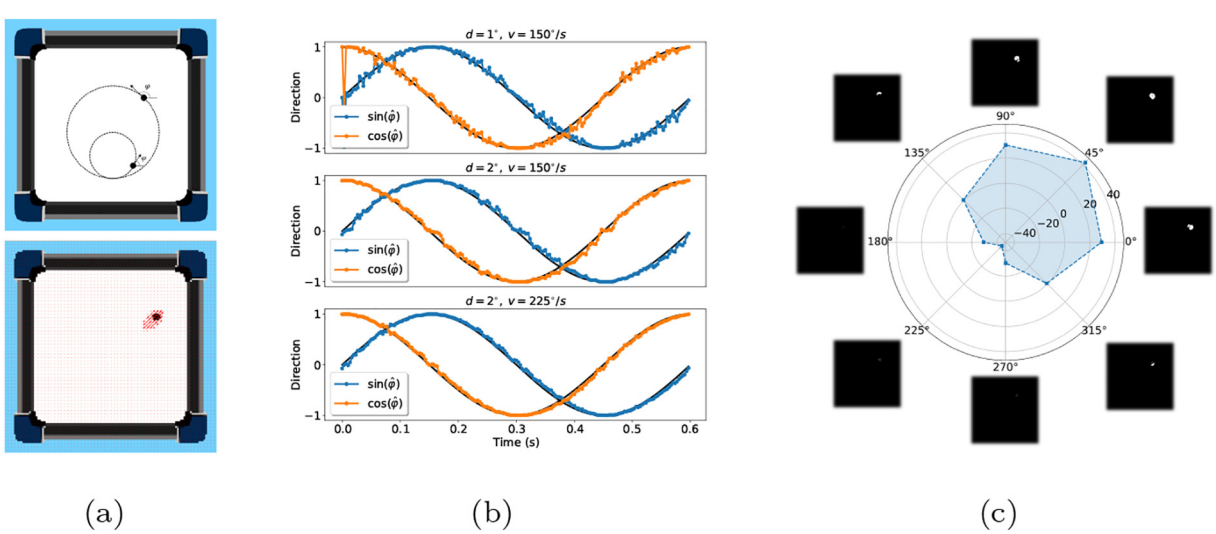

Fig. 5. Direction selectivity of the proposed BRNN. (a) Two cases of small-target direction perception. (b) Direction estimation at different velocities and target sizes (circular motion). (c) The polar diagrams of motion energy at different filtering orientations (45° motion).

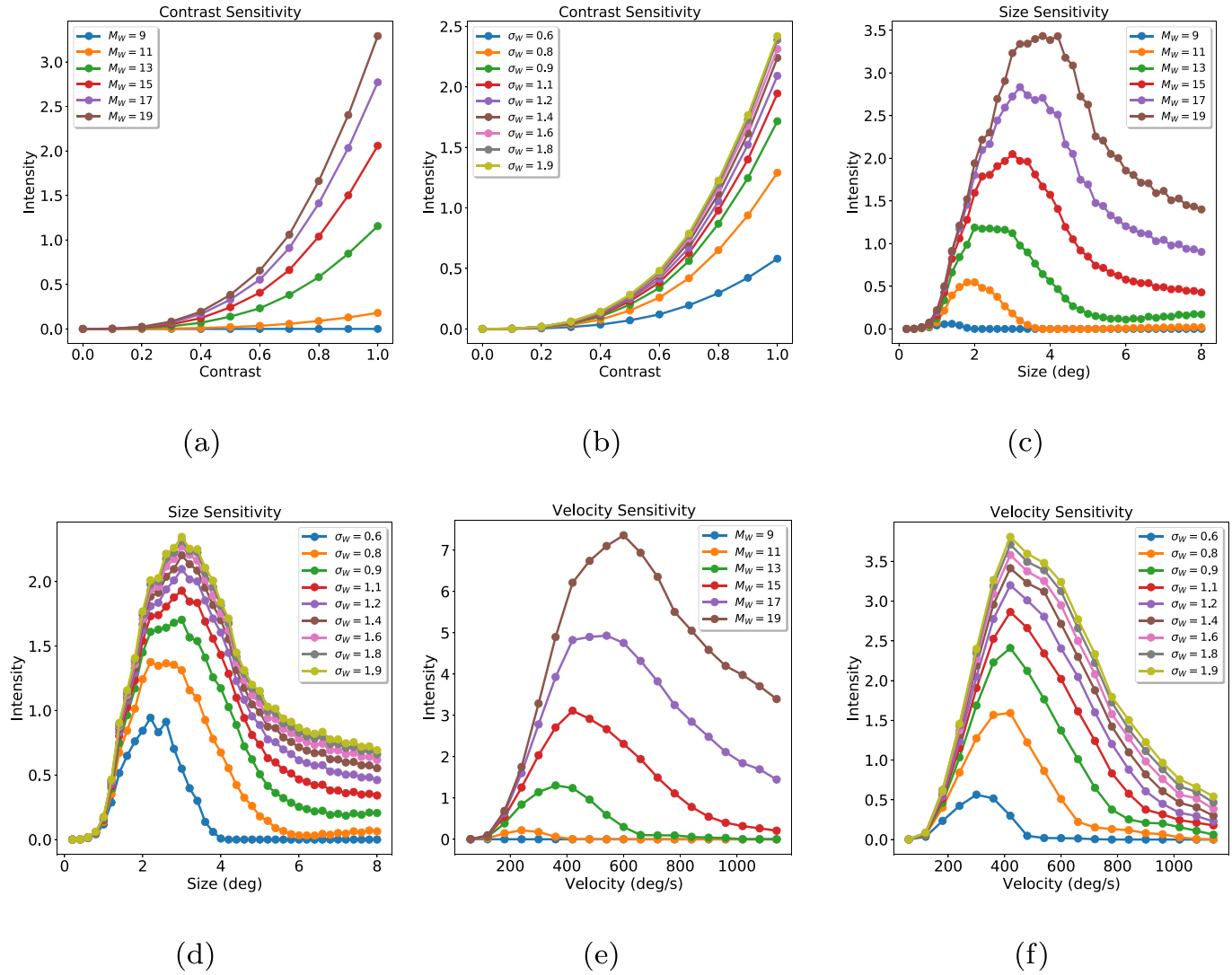

Fig. 6. The contrast, size and velocity sensitivity of the BRNN under different conditions. (a) The contrast tuning at different kernel size M_W ($\sigma_W = 1.2$). (b) The contrast tuning at different σ_W ($M_W = 15$). (c) The size tuning at different kernel size M_W ($\sigma_W = 1.2$). (d) The size tuning at different σ_W ($M_W = 15$). (e) The velocity tuning at different kernel size M_W ($\sigma_W = 1.2$). (f) The velocity tuning at different σ_W ($M_W = 15$).

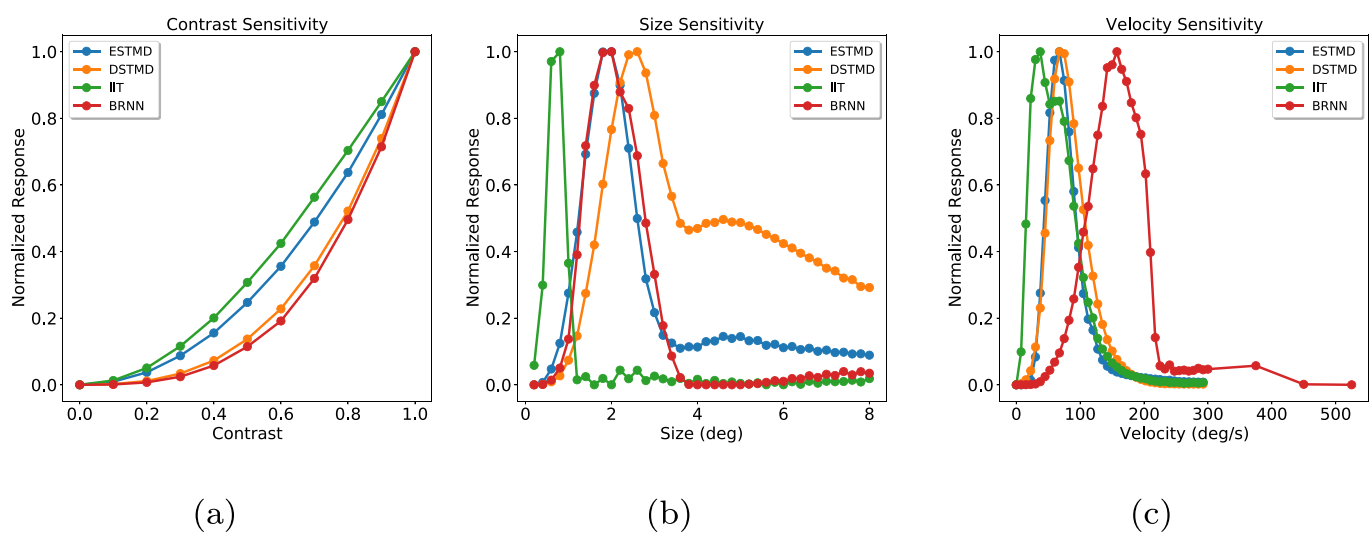

Fig. 7. The contrast, size and velocity sensitivity of the BRNN compared with other typical bioinspired models. (a) The normalized mean responses with different contrast. (b) The normalized mean responses with different sizes. (b) The normalized mean responses with different velocities.

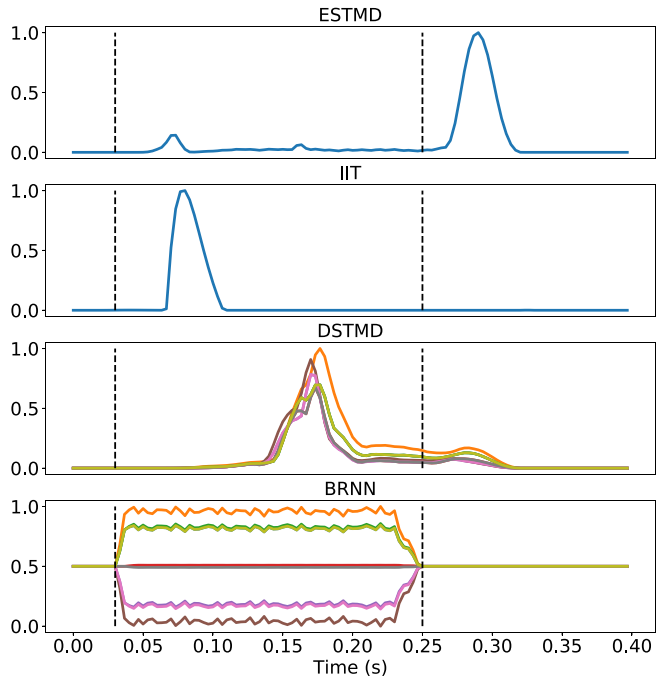

Fig. 8. The normalized outputs of four models at $d = 2^{\circ}$, $v = 75^{\circ}/s$.

$$\begin{aligned} V_{+}(x,y,t) &= \max_{\theta} E_{+}^{\Theta}(x,y,t)\cdot WA_{+}(x,y,t) \\ V_{-}(x,y,t) &= \max_{\theta} E_{-}^{\Theta}(x,y,t)\cdot WA_{-}(x,y,t) \end{aligned} \tag{29}$$

In the same way, the ON and OFF pathways are combined through the weighted summation.

$$V(x,y,t) = \left[w_{+}V_{+}(x,y,t) + w_{-}V_{-}(\mathrm{x,y,t})\right]^{+} \tag{30}$$

It is worth noting that the activation of RGC $V(x,y,t)$ synthesizes the information of motion energy and size of the moving object, which is jointly mediated by contrast, motion velocity, motion direction and target size. It is an ideal object for performing small-target motion detection.

Specifically, there are two steps to execute this procedure. Firstly, $Norm(V)$ are used to extract regions of interest, where $Norm(\cdot)$ is a Min-Max normalization function. For a given detection threshold γ, if $Norm(V(x,y,t)) > \gamma$, the location (x,y) is added to the regions of interest. Then the DBSCAN [44,45] algorithm is used to discover the clusters of excitatory neural populations. The clustering center (X,Y) is regarded as the position of the candidate target. The motion direction can be estimated by a population coded algorithm. The maximal motion energy of each point is decomposed into two parts along the x and y axis.

$$\begin{aligned} E_x(x,y,t) &= \max_{\theta} E^{\Theta}(x,y,t)\cos\left(\widehat{\varphi}(x,y,t)\right) \\ E_y(x,y,t) &= \max_{\theta} E^{\Theta}(x,y,t)\sin\left(\widehat{\varphi}(x,y,t)\right) \end{aligned} \tag{31}$$

Then the maximal motion energy of each cluster can be computed via average method.

$$\begin{aligned} E_X(x,y,t) &= \frac{1}{N_{\mathcal{T}}}\sum_{(x,y)\in\mathcal{T}} E_x(x,y,t) \\ E_Y(x,y,t) &= \frac{1}{N_{\mathcal{T}}}\sum_{(x,y)\in\mathcal{T}} E_y(x,y,t) \end{aligned} \tag{32}$$

where $\mathcal{T}$ is the point set of a candidate target, $N_{\mathcal{T}}$ is the number of points. The motion direction and energy of the moving target can be estimated as follows.

$$\begin{aligned} \Phi(X,Y,t) &= \arctan 2(E_X(x,y,t), E_Y(x,y,t)) \\ E(X,Y,t) &= \sqrt{E_X(x,y,t)^2 + E_Y(x,y,t)^2} \end{aligned} \tag{33}$$

In the real world, a moving background brings huge disturbance to motion detection since many of the features are similar to small moving targets in this situation. As one of key characteristics, these fake targets usually moves in the same direction when the camera shakes. In order to decrease this disturbance, we propose a background suppression method based on the algorithm of directionally selective inhibition, as shown in Fig. 4. The motion direction $(0 - 2\pi)$ is equally divided into eight intervals. The number of targets is then counted in each interval. If the number of targets is greater than a threshold N_{th} in the internal $[a,b]$, a and b are added to the set of inhibitory direction Θ_I. Then the activities of RGCs with directionally selective inhibition become the following form.

$$\begin{aligned} V'_{+}(x,y,t) &= V_{+}(x,y,t) - \max_{\theta_I} E_{+}^{\Theta_I}(x,y,t)\cdot WA_{+}(x,y,t) \\ V'_{-}(x,y,t) &= V_{-}(x,y,t) - \max_{\theta_I} E_{-}^{\Theta_I}(x,y,t)\cdot WA_{-}(x,y,t) \end{aligned} \tag{34}$$

Finally, $Norm(V')$ is used to determine the target location. In the same way, the location (x,y) is added to the region of interest if $Norm(V'(x,y,t)) > \gamma$. Then the DBSCAN algorithm is used to determine the position of targets.

4. Experimental results and discussion

In this section, extensive synthetic and real-data experiments are conducted to verify the the accuracy and effectiveness of the proposed algorithm. We compare the proposed model to some state-of-the-art

Fig. 9. The scene of multiple small targets moving in a cluttered background.

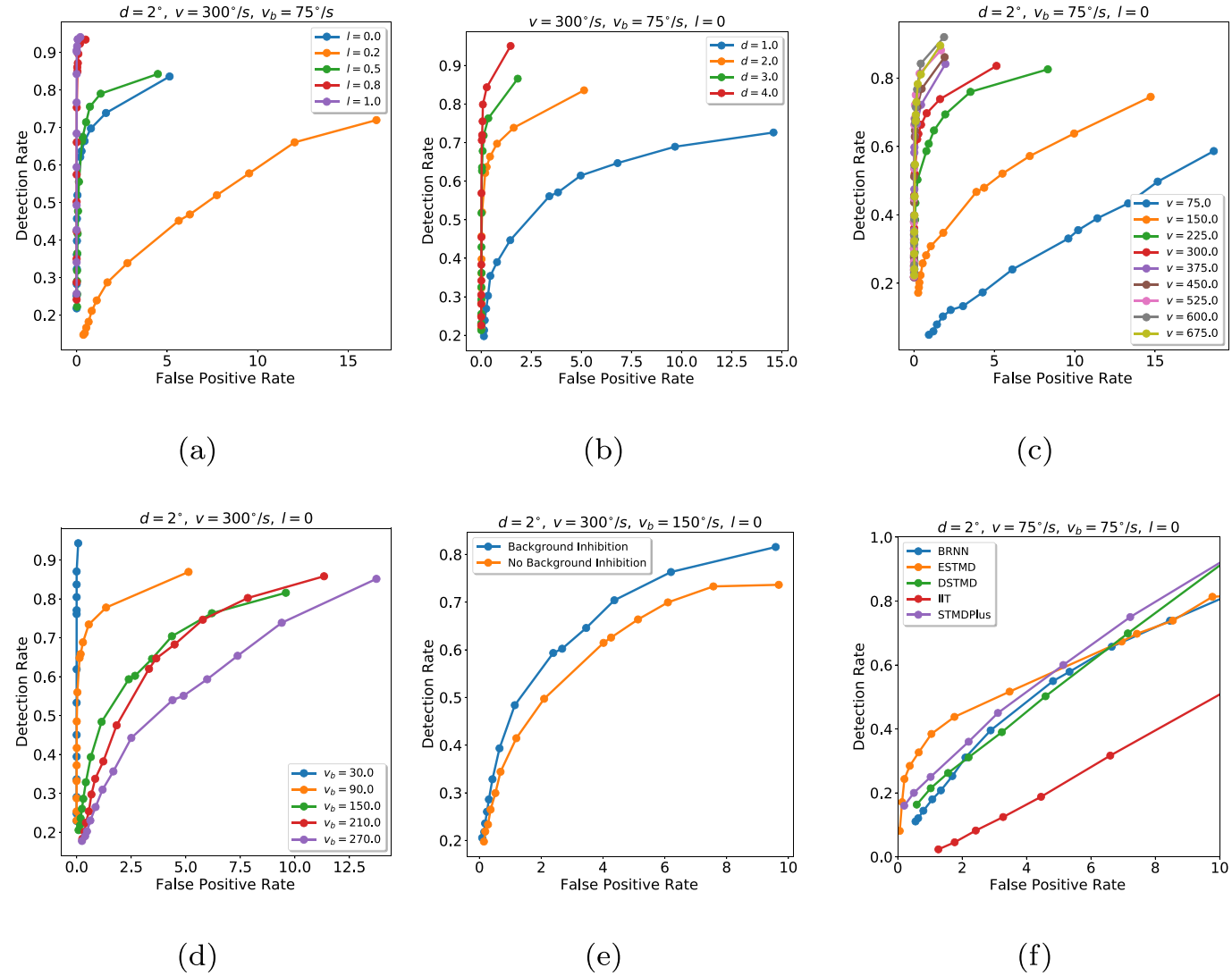

Fig. 10. The ROC curves of the BRNN for motion detection under different settings.

methods, including ESTMD [22], IIT [24], DSTMD [26] and STMDplus [27].

4.1. Basic properties of the proposed model

Based on the open source toolkit Psychlab developed by DeepMind, we firstly build a simple scene of small moving target, as shown in Fig. 5(a).

4.1.1. Direction selectivity

In the test of direction estimation, we let a small target move along a circular orbit. The input image is 128×128 gray-scale map and the parameters of model are listed in Table 1. The sizes of kernal function g_I, g_S and g_W are set as 5×5. Different target sizes and velocities are chosen to evaluate the performance of direction estimation. The results show the proposed model has a good selectivity of motion direction, as plotted in Fig. 5(b). The sine and cosine of the estimated direction can track the real motion direction accurately. In the case of 45° motion, we record sum of motion energy from DSGCs in Fig. 5(c). Obviously, the closer the filtering orientation and the motion direction is, the higher the intensity is.

4.1.2. Contrast sensitivity

Contrast sensitivity represents that the visual response generally increases with the increase of contrast. In this paper, Weber contrast is used to test the contrast sensitivity, which is defined as $c = \frac{|\bar{I}_t - \bar{I}_b|}{255}$. $\bar{I}_t$ is the average gray-scale value of moving target, and $\bar{I}_b$ is the average gray-scale value in neighboring area around the target. We compare the neural response under different kernel sizes M_W and standard deviations σ_W. The response intensity is plotted in Fig. 6(a) and Fig. 6(b). As the contrast increases, the outputs of our model grow approximately quadratically. Meanwhile, the larger M_W and σ_W generate a larger excitation/inhibition ratio, which causes the increase of response intensity. Furthermore, we compare the contrast sensitivity of BRNN with other typical bioinspired models, including ESTMD [22], IIT [24] and DSTMD [26]. The normalized responses of them are similar, as shown in Fig. 7(a). In addition, the STMDplus has the same basic structure as the DSTMD model [27] such that their basic properties are the same.

4.1.3. Size and velocity sensitivity

Size and velocity sensitivity are basic properties of motion vision [46]. Fig. 6(c) and Fig. 6(d) show the size tuning with different M_W and σ_W. Obviously, the response amplitude reaches the peak at an

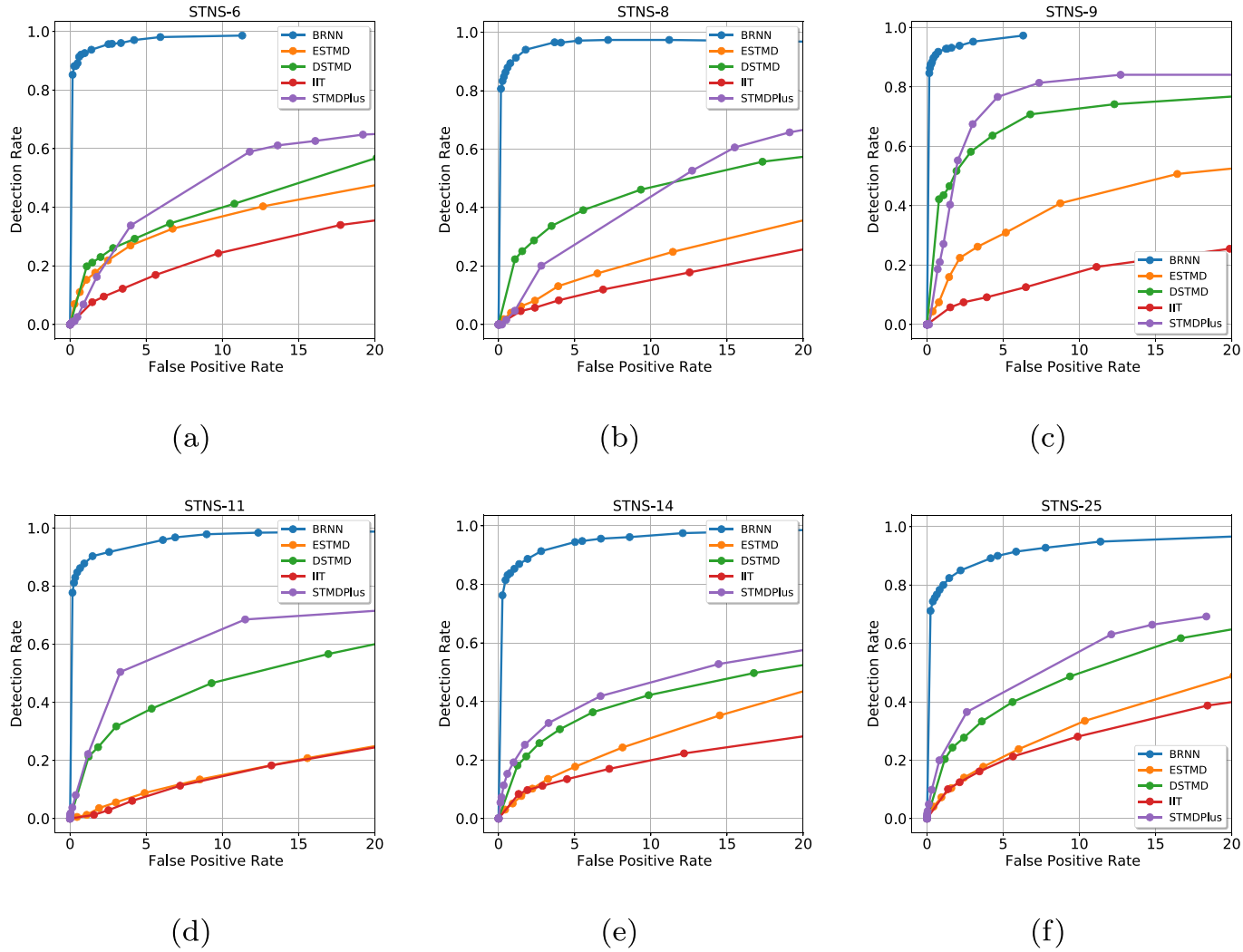

Fig. 11. The ROC curves of five bioinspired models for small-target motion detection in different STNS datasets.

optimal size ($2^{\circ} \sim 4^{\circ}$), which is consistent with the biological data [41]. The increased kernel size and standard deviation can enhance response amplitude and enlarge the optimal size. One explanation is that the optimal size closely matches with the dendrite field size [41], which is positively related to M_W and σ_W. The size tuning of BRNN, ESTMD, IIT and DSTMD (STMDplus) are shown in Fig. 7(b). By adjusting $M_W = 11$ and $\sigma_W = 1.2$, BRNN shows the size tuning similar to the ESTMD.

Fig. 6(e) and (f) show the velocity tuning with different M_W and σ_W. The velocity sensitivity obeys similar variation rule to the size sensitivity. The increased M_W and σ_W can enhance response amplitude and enlarge the optimal velocity as well. Additionally, the optimal velocities of other models are generally lower ($50^{\circ} \sim 100^{\circ}/s$), as shown in Fig. 7(c). While BRNN has a larger range of velocity sensitivity by changing M_W and σ_W.

4.1.4. Response speed

The typical temporal band-pass filter usually suffers from phase delay in discrete digital filtering process, which has a strong impact on accuracy and realtime performance. In this paper, the proposed neurodynamics-based temporal filter can achieve the same function as the traditional band-pass filter in essence. But it is easy to reduce phase delay through selecting the superficial neural outputs, while without affecting the filtering performance. The normalized outputs of four models are plotted in Fig. 8, where a small target moves at a constant speed from $0.03 - 0.25s$. The ESTMD, IIT and DSTMD (STMDplus) have distinct phase delay, and the response time is about $40ms$. While the proposed temporal filtering enables the model to respond to motion rapidly.

4.2. Motion detection in a synthetic scenario

In this section, the ability of motion detection in cluttered backgrounds is tested. The process of motion detection is described in Section 3.4. A receiver operating characteristic (ROC) curve is plotted to indicate the diagnostic ability. This curve is created by plotting the true-positive rate (TPR) against the false-positive rate (FPR) at various threshold γ settings. The true-positive rate represents the probability of successful detection, which is calculated by the number of true detections N_{TD} and the number of actual targets N_{AT}.

$$TPR = \frac{N_{TD}}{N_{AT}} \tag{35}$$

The false-positive rate represents the probability of false alarm. The total number of time steps is fixed at N_T, we set the number of false positives FPR as abscissa, defined as follows.

Table 2
The detection rate of five models at $FPR = 5$.

Dataset	ESTMD	IIT	DSTMD	STMDplus	BRNN
STNS-1	0.11	0.16	0.34	0.41	**0.66**
STNS-2	0.20	0.03	0.41	0.53	**0.68**
STNS-3	0.10	0.12	0.16	0.33	**0.96**
STNS-4	0.09	0.08	0.12	0.26	**0.89**
STNS-5	0.12	0.07	0.48	0.56	**0.97**
STNS-6	0.29	0.16	0.31	0.42	**0.98**
STNS-7	0.06	0.06	0.12	0.49	**0.88**
STNS-8	0.15	0.10	0.38	0.40	**0.97**
STNS-9	0.31	0.11	0.66	0.79	**0.96**
STNS-10	0.17	0.04	0.44	0.54	**0.82**
STNS-11	0.09	0.08	0.37	0.73	**0.95**
STNS-12	0.27	0.22	0.43	0.49	**0.81**
STNS-13	0.31	0.29	0.46	0.41	**0.48**
STNS-14	0.18	0.14	0.33	0.37	**0.94**
STNS-15	0.24	0.11	0.28	0.33	**0.74**
STNS-16	0.11	0.05	0.14	0.37	**0.66**
STNS-17	0.17	0.08	0.11	0.16	**0.64**
STNS-18	0.34	0.31	0.38	0.45	**0.58**
STNS-19	0.21	0.19	0.26	0.46	**0.84**
STNS-20	0.25	0.15	0.35	0.37	**0.70**
STNS-21	0.15	0.23	0.27	0.43	**0.74**
STNS-22	0.58	0.59	**0.88**	0.79	0.81
STNS-23	0.38	0.48	0.54	0.62	**0.77**
STNS-24	0.11	0.26	0.23	0.42	**0.61**
STNS-25	0.21	0.20	0.38	0.58	**0.91**
Avg.	0.21	0.17	0.35	0.47	**0.80**

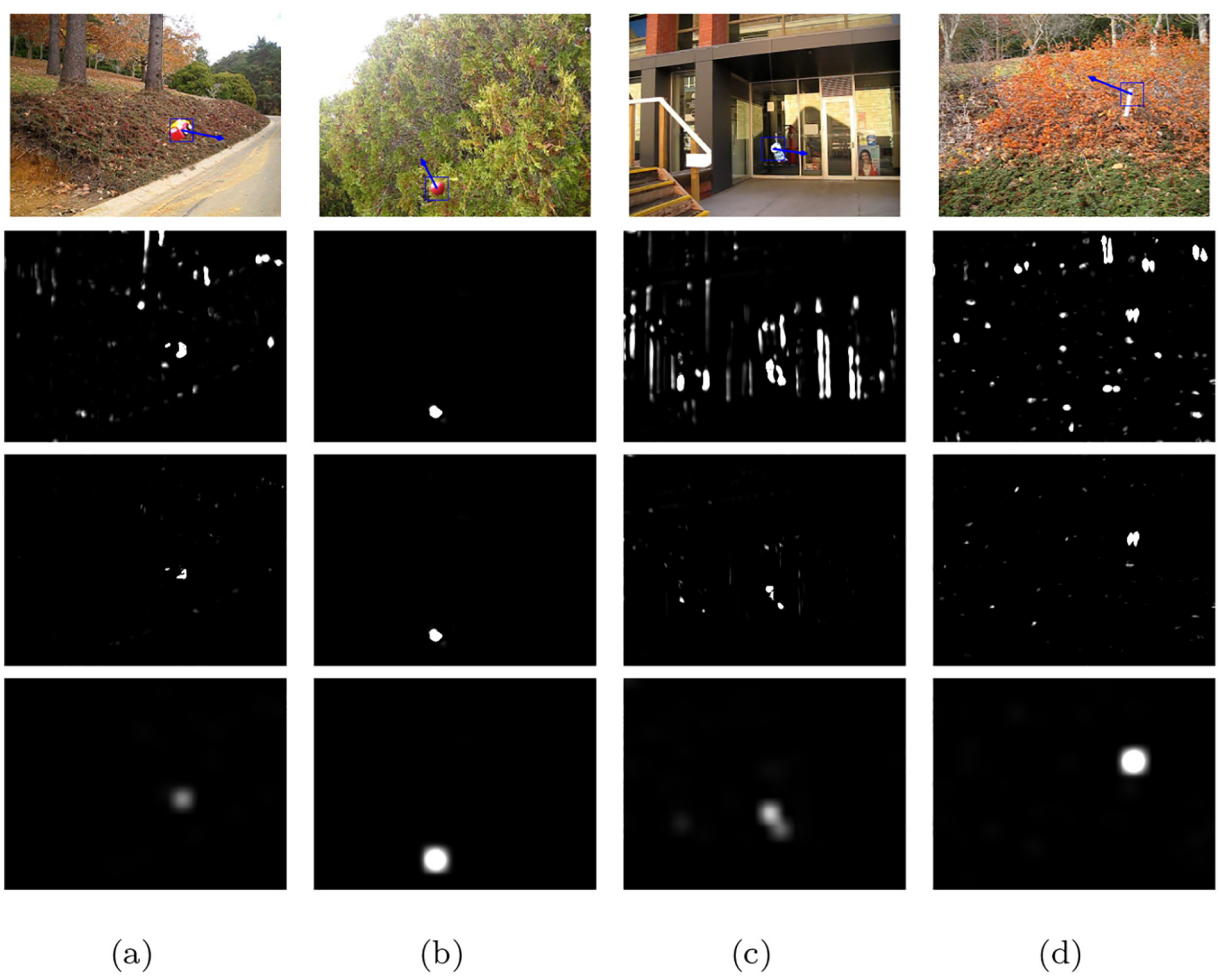

Fig. 12. Neural responses of the proposed BRNN. Line 1: the position and motion direction of the small moving target. Line 2: the activities of DSGCs V. Line 3: the activities of DSGCs V' with directionally selective inhibition. Line 4: The responses of V' after Gaussian filtering.

Table 3
The computing time of four models for each step.

ESTMD	IIT	DSTMD	BRNN
0.065 ± 0.004	0.032 ± 0.034	0.085 ± 0.004	0.084 ± 0.003

$$FPR = \frac{N_{FP}}{N_T} \tag{36}$$

The synthetic environment is displayed in Fig. 9, where the background is cluttered and dark. The cluttered backgrounds moves from left to right at different speed. Five targets move from random initial positions in five random directions respectively. Six groups of experiments are conducted to investigate the influences of luminance, size, velocity of targets, velocity of background and background suppression algorithm. The luminance, diameter and velocity of targets are set to $l = 0$, $d = 2^{\circ}$ and $v = 300^{\circ}/s$ respectively. The velocity of background is in default set to $v_b = 75^{\circ}/s$. We use the DBSCAN algorithm in scikit-learn to discover the clusters of the regions of interest. If the distance between the clustering center and the actual target position less than a threshold $d_{th} = 0.5d + 1^{\circ}$, the target is considered to be detected successfully.

The diagnostic ability of the proposed motion detector is firstly evaluated with different target luminance. The luminance of targets is set to $I = 0.0, 0.2, 0.5, 0.8, 1.0$ respectively, correspongding to the actual RGB value $[255I, 255I, 255I]$. The ROC curve is plotted in Fig. 10(a). It is clear that BRNN has a good performance when targets are very dark or bright, while the performance is lower when the luminance is 0.2. This is because the luminance of background is close to 0.2 such that the Weber contrast is low when $l = 0.2$.

The Fig. 10(b) and (c) show the detection performance of BRNN with different target sizes and motion velocities. It is effective to detect small targets around the optimal size and velocity. As the size and velocity increase, the detection rate of BRNN also increases for a given false positives. Meanwhile, the optimal size can be adjusted by changing the parameters of antagonistic center-surround spatial filtering, which can expand the range of size and velocity sensitivity.

The Fig. 10(d) reflects the influence of background velocity on small-target motion detection. Fast background movement can weaken the capability of detection since a faster moving background produces many small target-like features and increases false positive rate greatly. The directionally selective inhibitioin algorithm can effectively reduce the interference from the moving background in our model, which is reflected in Fig. 10(e). This is significant for robotic visual system to overcome the target detecting problem with an instable visual input.

Finally, we also compare the performance of our BRNN with ESTMD, DSTMD, STMDplus and IIT at the optimal value of these classical models, where $d = 2^{\circ}$, $v = 75^{\circ}/s$. We set the kernel size and standard deviation to $M_W = 11$, $\sigma_W = 0.7$. The ROC curve is plotted in Fig. 10(f). The ESTMD, DSTMD and STMDplus are relatively good at detecting slow motion. The proposed BRNN is also able to obtain similar detection performance but without competitive advantage. However, it is worth noting that sensing slow movement usually needs high sampling frequency, which may be difficult to perform in low-cost hardware systems.

4.3. Motion detection in real scenarios

The proposed model is further tested in the STNS dataset [24], which contains small moving targets within several real cluttered natural environments. All frames are used to evaluate the diagnostic ability of the proposed motion detector. All images are resized to one-half as 320 × 240. The FOV for this dataset is also not available, therefore we assume FOV ~80°. Additionally, the distance threshold d_{th} is set to $0.5d + 1^{\circ}$, where d represents the diagonal length of the target. For each scenario, the detection threshold γ is chosen as $0.01 - 0.09$ (step = 0.01) and $0.1 - 0.9$ (step = 0.1), which is used for plotting ROC curve. The parameters of the antagonistic center-surround spatial filter are set to $M_W = 21$, $\sigma_W = 1.5$.

We compare the results of ESTMD, DSTMD, STMDplus, IIT and our proposed BRNN at the same experimental conditions. The ROC curves of STNS-6, 8, 9, 11, 14 and 25 are respectively plotted in Fig. 11, where the areas under the curve (AUC) of BRNN are clearly greater than other models. The ROC curves of all datasets are placed in the supplementary material. We list the detection rate of all scenarios at $FPR = 5$ in Table 2. Obviously, the proposed model works accurately and stably for all different targets and cluttered backgrounds. Additionally, STMDplus model outperforms DSTMD, ESTMD and IIT, which is consistent with the results in [27].

This mainly benefits from two reasons. Firstly, the novel spatiotemporal energy model has a wider size and velocity sensitivity range such that it is effective to capture all possible targets. Secondly, this model can estimate motion direction accurately, which enables the directionally selective inhibition method to suppress the target-like features in the background successfully. Thus, the fake targets are removed effectively from the possible targets such that the detection rate increases and the false positives decreases. Taking STNS-6, 9, 23 and 25 for example, we plot the activities of DSGCs V, DSGCs with directionally selective inhibition V' and V' after Gaussian filtering respectively. As shown in Fig. 12, BRNN can not only accurately detect the position of small moving targets, but also extract their motion direction and motion energy at the same time. Meanwhile, the algorithm of directionally selective inhibition can suppress the activations caused by the moving background.

Finally, We further examine the processing speed of different models. All experiments are conducted in a computer, which has an ADM Ryzen 53,600X 6-core 3.8GHz Processor and 16GB of RAM. It runs an Ubuntu 18.04 LTS 64-bit operating system. The architecture has been developed in Python 3.8 and OpenCV. The computing time is mainly related to the size of the input image and the complexity of model. With respect to the 320 × 240 input image, the computing time for each step is recorded in Table 3. The IIT model has the fastest processing speed due to its relatively simple and optimized spatiotemporal filtering process. While the computing cost of the proposed BRNN is close to the DSTMD, since both of them use directionally selective spatial filtering to estimate motion direction and adopt the DBSCAN algorithm to select the region of interest. In addition, the time and space complexity of STMDplus is higher than the DSTMD. Although they have the same main structure, the trajectory of each small feature point needs to be recorded to determine whether it is a small target in the STMDplus model. The computing time of this model depends on the number of small feature points.

5. Conclusion

It is significant and challenging for robotic visual systems to improve the diagnostic ability of small target motion detection in cluttered moving backgrounds. Inspired by elementary motion vision in the mammalian retina, this paper presented a BRNN that can not only detect small moving targets in the cluttered moving background, but also extract the information of motion direction and motion energy. BRNN is a novel spatiotemporal energy model based on a neurodynamics-based temporal filtering and 2-D spatial Gabor filtering, simulating the basic functions of photoreceptor cells, bipolar cells, amacrine cells, and ganglion cells in the retina. It is able to estimate motion direction accurately and respond to small targets of different sizes and velocities through changing the dendrite field size of spatial filter. In addition, an algorithm of directionally selective inhibition is proposed to suppress the target-like features in the moving background, which can reduce the influence of background motion effectively. Comparing with other bioinspired models, the proposed model works stably for small targets of a wider size and

velocity range, and has better detection performance in the synthetic and real data experiments. The current research provides an attractive alternative paradigm for robotic search and tracking applications. It is also very meaningful for the research of biology and computational neuroscience.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported in part by the National Key Research and Development Program of China under 2018YFB1305300, the China Postdoctoral Science Foundation under Grant 2020TQ0039, the National Natural Science Foundation of China under Grant 61733001, U2013602, 61873039, U1913211 and U1713215.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.imavis.2021.104266.

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